Design Requirements Analysis – Team 2

School of Engineering · MSc Robotics · AERO62520 Robotic Systems Design Project

Section 1 – Problem Statement

The stakeholders need a robot that can autonomously navigate an environment having flat surfaces and obstacles to identify various colored objects. Once identified, the robot needs to pick and transport each object to a designated bin that matches its color and deposits it there. This process should be fully automated, with the robot continuously searching for colored objects and sorting them into the correct bins based on color recognition, without any human assistance.

Section 2 – Concept of Operations

1. **System Initialization**

   The robot is placed at the starting point within the arena. Once the operator presses the start button, all sensors and control subsystems (camera, LiDAR, processors, motors, and gripper) are activated.

2. **Environment Scanning and Mapping**

   The robot scans its surroundings using the camera and LiDAR to detect obstacles, colored objects, and bins. It then generates or updates a map of the environment for subsequent navigation and planning.

3. **Target Search and Path Planning**

   The robot analyses sensor data to determine which target object should be retrieved first. After that, compute an optimal path to reach the target bins while avoiding obstacles.

4. **Object Recognition and Localization**

   The robot stops near the target and performs a detailed scan. The vision system identifies the object's color and determines an appropriate grasping pose.

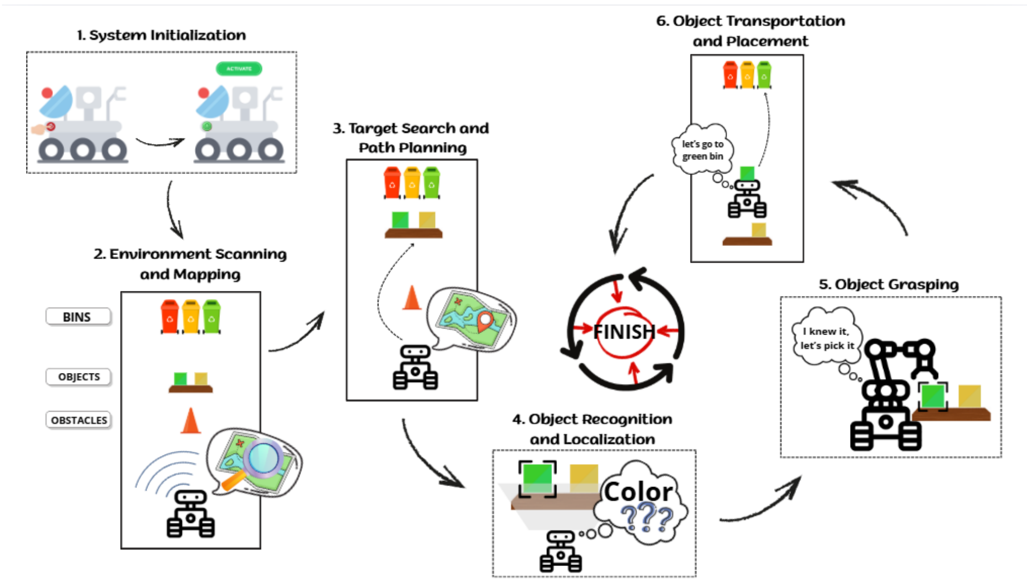
5. **Object Grasping**

   The robotic gripper executes a controlled grasp to pick up the identified object securely. Once the object is firmly held, the robot transitions to navigation mode.

6. **Object Transportation and Placement**

   The robot plans a path toward the corresponding bin based on the object's color, moves toward the target while avoiding obstacles, and safely deposits the object inside the correct bin.

Concept of Operations (CONOPS) Flow



Section 3 – Functional and Performance Requirements

Require-ment ID	Requirement	Source/s	Assumptions and Reasoning
1	**Functional**		
1.1 Motion & Detection	**The robot should move around and detect colored objects and bins using on-board sensors.**	**user requirements**	**The objects located between 0.3 and 3 meters away from the camera inside the arena would be detected.**
1.1.1	The robot shall have camera	team discussions	usage of Intel RealSense Depth Camera
1.1.2	The robot shall have lidar	team discussions	usage of RPLidar A2M8/A2M12
1.1.3	The robot shall have a computation device	team discussions	usage of Intel NUC
1.1.4	The robot shall have actuators to move	team discussions	usage of wheel to move
1.2 Navigation	**The robot should navigate autonomously.**	**user requirements**	**The system has been built and set-up procedurally**
1.2.1	The robot shall target objects while avoiding obstacles and boundaries.	team discussions	robot will do navigation using SLAM
1.3 Grasping	**The robot should grasp, lift, and place the objects inside the identified bin.**	**user requirements**	**all those components are in the arena, and positioned in the proper way to be picked up**
1.3.1	The robot shall have gripper as end effector	team discussions	usage of adaptive gripper from mycobot
1.4 Classification	**The robot should classify each detected object and bin**	**user requirements**	**robot only needs to pick the recognized object**
1.4.1	The classification process shall be done using Specific Computer Vision method	team discussions	usage of Open CV method to detect object and distinguish color classes
1.4.2	The picked objects and bins shall match with specific class of color (yellow or blue)	team discussions	if there are other color objects and bins, those will be ignored
2	**Enviromental**		
2.1 Surfaces	**The arena will have surfaces**	**site visit**	**The robot can move freely on the surface.**
2.1.1	the surface will be flat and level	site visit	Wheels do not move just because gravity
2.1.2	The surface will have static friction and do not provide additional adhesive force (adhesion) other than normal friction.	site visit	Wheels can roll
2.2 Brightness	**The arena will have lighting**	**site visit**	-
2.2.1	Brightness level will be between 300-800 lx	environment analysis	The arena has indoor lighting
2.3 Temperature	**The arena's temperature will be room temperature**	**site visit**	-
2.3.1	The temperature shall be between 0 to 40 deg Celsius	environment analysis	The room's temperature suitable for each component of robot
2.4 Dimension	**The arena will have static dimension**	**site visit**	**There is no boundaries movement**
2.4.1	The arena's width and length shall not exceed 23.8 meters.	devices specifications	The boundaries of arena can be detected by Lidar
2.5 Picked Object	**The arena will have objects to be picked**	**user requirements**	**-**

Require-ment ID	Requirement	Source/s	Assumptions and Reasoning
2.5.1	Objects shall reflect light in the yellow wavelength range (≈570–590 nm) or violet–red wavelength range (≈380–450 nm mixed with ≈620–750 nm)	team discussions	The objects that will be used are the same with the given kit
2.5.2	The picked objects shall be normally static	user requirements	The position of the picked object can be mapped.
2.5.3	The objects shall have at least two parallel sides distancing no more than 35mm	team discussions	The objects have side that can be gripped using adaptive gripper
2.5.4	The object will be solid, has friction and not adhesive	team discussions	objects surfaces are not sticky and slippery
2.6 Obstacles	**The arena will have obstacles**	**user requirements**	**-**
2.6.1	obstacles shall be static	user requirements	The position of the obstacles can be mapped.
2.6.2	The height of obstacles shall be greater than the lidar position from the ground	system analysis	the obstacles can be scanned by lidar
2.6.3	The obstacles shall not block the entire way	system analysis	there is space for robot to move
2.7 Bins	**The arena will have bins**	**user requirements**	**-**
2.7.1	the bins shall be static	user requirements	The position of the bins can be mapped.
2.7.2	The provided bins' color shall be at least the same as the picked objects	user requirements	the bin size can store the objects and suitable for the robotic arm
3	**Maintainability**		
3.1 Charging	**The robot's battery will need to be charge**	**system requirements**	**-**
3.1.1	the battery will be chargeable	system requirements	the battery and charger are suitable
3.2 Tightening	**Fix mechanical junctions will need to be tightened if loose**	**system requirements**	**-**
3.2.1	All the screws, nuts and bolts have appropriate sizes	system requirements	All the screws, nuts and bolts are tight enough to hold the structure
4	**Operability**		
4.1 Timing	**The robot should work for specific time**	**user requirements**	**-**
4.1.1	Operator will prepare the robot within 5 mins	user requirements	the robot has been tested to do the task
4.1.2	The robot shall do the required tasks within 20 mins	user requirements	the preparation process does not have any error
4.2 Robot's Status	**The robot may give status**	**team discussions**	**-**
4.2.1	The robot may give status during the process through CLI or GUI	team discussions	to track the progress of the process
4.3 Error Handling	**The robot may perform error during the process**	**system requirements**	**-**
4.3.1	User may interrupt process when recognizes error	team discussions	-
4.4 Safety	**The operator shall mitigate emergency condition**	**team discussions**	**-**
4.4.1	The robot shall include emergency stop button	team discussions	-

Require-ment ID	Requirement	Source/s	Assumptions and Reasoning
4.5 Manual control	**The robot shall be able to be controlled manually**	**user requirements**	**-**
4.5.1	The joystick shall be capable to control the robot's movement	team discussions	in case of malfunction or necessary scenarios if any
5	**Mechanical/structural**		
5.1 Structure	**The robot shall have chassis to support system integration and movement**	**system analysis**	**-**
5.1.1	The robot's chassis can support a payload of up to 3 kg	system analysis	The robot only carries an arm, depth camera, lidar, and NUC.
5.1.2	The robot will use wheels to navigate.	system requirements	The project will use Leo Rover robot
6	**Electrical**		
6.1 Power Supply	**The robot will use battery or power supply to operate**	**system requirements**	**-**
6.1.1	The battery shall have a 12V DC output	devices specifications	voltage conversion is allowed
6.1.2	The battery shall have minimum runtime 30 minutes	system requirements	work under typical load

Section 4 – Requirements Verification Matrix

Requi-rement No.	Source	Shall Statement	Verification Success Criteria	Respon-sible Party?	Agreed Results / Output of test
1.1	Requirements – Functional	The robot shall autonomously patrol the 2 m × 2 m arena at ≤ 0.3 m/s, maintain a local map, and detect colored objects and bins in three distinct zones without manual aid.	1) Place objects and bins in three zones; start mission and record path/detections. 2) In one patrol, per-class precision/recall ≥ 0.8 (or ≥ 8/10 true detections); false positives ≤ 2. 3) No collisions; no boundary crossings; trajectory file saved.	Team	Video of patrol; detection log; path/ros-bag file.
1.1.1	Requirements – Functional	The robot shall include a forward-facing RGB camera and publish '/camera/image_raw' at ≥ 10 Hz with resolution ≥ 640Ö480 calibration loads on startup.	1) Record 60 s rosbag; measured topic rate ≥ 10 Hz and frames at ≥ 640Ö480. 2) Startup log shows the calibration file loaded without errors. 3) Spot-check 10 frames — no tearing or major drops.	Test Eng.	Topic rate screenshot; 60 s bag; calibration file/log.

Requirement No.	Source	Shall Statement	Verification Success Criteria	Responsible Party?	Agreed Results / Output of test
1.1.2	Requirements – Functional	The robot shall include a 2D LiDAR that publishes '/scan' at $\geq$ 5Hz with an effective indoor range of $\approx$ 0.15–3.5 m for mapping and obstacle detection.	1) Display '/scan' for 60 s; average rate $\geq$5 Hz with no timeouts. 2) Validate ranges at 1 m and 2 m against a ruler/board (error $\leq\pm$10 cm). 3) Generate an occupancy grid in a static layout and visually match obstacles.	Test Eng.	Scan screenshot; occupancy grid capture; 60 s bag.
1.1.3	Requirements – Functional	The robot shall include an on-board computation device capable of launching perception, control, and teleoperation nodes via a single command and running for $\geq$5 minutes without crash.	1) Launch all nodes with one command; all required topics/services visible. 2) Operate 5 min with no crash; average CPU <90%, no thermal throttling. 3) Power remains stable; logs show no critical errors.	Test Eng.	Launch log; node/topic list;
1.1.4	Requirements – Functional	The robot shall actuate its wheels to drive straight 2 m ($\pm$0.2 m) and perform a 90° ($\pm$10°) turn on command, then stop within 0.2 m.	1) Follow taped 2 m straight line; end error $\leq\pm$0.2 m. 2) Execute 90° turn; angle error $\leq\pm$10°. 3) From stop command, final overrun $\leq$0.2 m; no manual correction.	Test Eng.	Video with floor marks; odom/IMU log; timing sheet.
1.2	Requirements – Functional	The robot shall autonomously navigate from the start area to the object area, then to the bin area and back to start without collisions or boundary violations.	1) Plan and execute the complete route once; record the path. 2) Zero collisions and zero boundary crossings; minimum clearance $\geq$5 cm. 3) Save navigation logs and trajectory for replay.	Team	Navigation video; trajectory/rosbag; incident sheet (0/0).
1.2.1	Requirements – Functional	The robot shall approach selected objects while avoiding obstacles and arena boundaries, maintaining safe clearance at all times.	1) Ten approach trials; $\geq$8 without touching obstacles. 2) Boundary violations = 0; minimum clearance $\geq$5 cm. 3) Log approach distance and success/fail per trial.	Team	Video of 10 trials; approach distance log.
1.3	Requirements – Functional	The robot shall grasp, lift, and place target objects fully inside the identified bin without bouncing out.	1) Ten pick-and-place attempts; $\geq$8 objects end fully inside the chosen bin. 2) No rim collisions observed; no bounce-outs. 3) Record attempt time and outcome tally.	Team	Video of 10 attempts; tally sheet; timestamps.

Requirement No.	Source	Shall Statement	Verification Success Criteria	Responsible Party?	Agreed Results / Output of test
1.3.1	Requirements – Functional	The robot shall include a gripper end-effector that can open/close within 1.0 s and grasp the 35 mm test object.	1) Cycle open/close three times; each $\leq$1.0 s (timer visible). 2) Grasp 35 mm test piece 3/3 and lift $\geq$5 cm. 3) Release resets to neutral without sticking.	Test Eng.	Video; timing sheet; test object photo.
1.4	Requirements – Functional	The robot shall classify each detected object and bin into color classes (yellow/blue) with accuracy $\geq$90% on a labeled lab set.	1) Evaluate with $\geq$50 frames per class; overall accuracy $\geq$90%. 2) Average per-frame decision latency $\leq$100 ms. 3) Export confusion matrix and freeze config version.	Test Eng.	Confusion matrix; dataset note; config/commit ID.
1.4.1	Requirements – Functional	The classification process shall use the specified computer vision pipeline as defined in the system configuration.	1) Configuration file shows the specified pipeline enabled; record version/commit. 2) Unit test on $\geq$20 sample images pass with expected outputs. 3) Run log contains no fatal errors.	Test Eng.	Config snippet; unit-test report; run log.
1.4.2	Requirements – Functional	Picked objects and destination bins shall match the specified color classes (yellow/blue).	1) Place 10 objects; $\geq$9 go into the correct color bin. 2) In a single end-to-end run, color mismatches = 0. 3) Record results and any misclassification notes.	Team	Video; placement tally; run summary.
2.1.1	Requirements – Environmental	The arena surface shall be flat and level with tilt $\leq$1° over 1 m in both X and Y directions.	1) Measure with bubble levels in X and Y over 1 m; readings $\leq$1°. 2) Inspect the path for steps/ramps; none present. 3) Photograph measurements for records.	Lab Tech	Photos of level readings; checklist entry.
2.1.2	Requirements – Environmental	The arena surface shall provide normal friction and shall not provide additional adhesive forces beyond typical static friction.	1) Tilt test with standard block: sliding begins at $\geq$15°. 2) White paper wipe: no sticky residue remains. 3) Record tilt angle and result.	Lab Tech	Tilt test photo; friction check note.
2.2.1	Requirements – Environmental	The arena shall provide lighting at 300–800 lx measured at robot height.	1) Measure lux at robot height at three locations; all within 300–800 lx. 2) Variation across the test area $\leq\pm$20%. 3) Store meter photos/readings.	Lab Tech	Lux-meter photos; reading sheet.

Requirement No.	Source	Shall Statement	Verification Success Criteria	Responsible Party?	Agreed Results / Output of test
2.3.1	Requirements – Environmental	The ambient temperature during tests shall be between 0–40 °C.	1) Measure ambient temperature before each run; within 0–40 °C. 2) If out of range, postpone the run. 3) Record value on pre-run checklist.	Lab Tech	Thermometer photo; checklist entry.
2.4.1	Requirements – Environmental	The arena's width and length shall not exceed 23.8 m, and the layout shall remain unchanged during testing.	1) Tape-measure width and length; both $\leq$23.8 m. 2) Mark layout; re-check after tests that positions are unchanged. 3) Archive photos/notes.	Lab Tech	Tape photos; layout photo; checklist.
2.5.1	Requirements – Environmental	The arena shall include objects to be picked that reflect within the specified color bands (yellow 570–590 nm violet-red mix 380–450 + 620–750 nm) corresponding to classes.	1) Verify color with standard cards/specs; class labels match. 2) Attach supplier spec or colorimeter reading if available. 3) Keep a class list of IDs.	Lab Tech	Photo vs. color card; supplier/spec note.
2.5.2	Requirements – Environmental	Picked objects shall be stationary at the beginning of each trial.	1) Confirm no motion before each trial; document on checklist. 2) No external force applied during runs.	Operator	Signed checklist; setup photo.
2.5.3	Requirements – Environmental	Each target object shall have at least two parallel sides with a distance of $\leq$35 mm to allow stable grasping.	1) Measure with calipers; distance $\leq$35 mm. 2) Sample $\geq$3 objects; record values.	Test Eng.	Caliper photos; measurement sheet.
2.6.2	Requirements – Environmental	The arena shall include obstacles	1) Install $\geq$2 obstacles; mark positions; light push does not move. 2) Measure obstacle top $\geq$ LiDAR center height + 20 mm (ruler). 3) Obstacles appear in scans at a typical range. Obstacles shall be fixed/static, and the top of each obstacle shall be at least 20 mm higher than the LiDAR center height.	Lab Tech	Photos with ruler; pre/post position photos; scan capture.
2.6.3	Requirements – Environmental	Obstacles shall not block all feasible routes	at least one continuous path from start to bins shall exist.1) Conduct a dry-run (no objects) to validate a continuous path. 2) Keep route map; ensure path remains open during tests.	Lab Tech	Layout photo; dry-run video; checklist.

Requirement No.	Source	Shall Statement	Verification Success Criteria	Responsible Party?	Agreed Results / Output of test
2.7	Requirements – Environmental	Two bins shall be present and secured	bins shall remain static, and their colors shall match the defined target classes.1) Fix bins to floor/table; touch displacement <5 mm. 2) Verify colors against color cards; mismatches = 0. 3) Positions are unchanged after running.	Lab Tech	Bin photos; color-card checklist.
3.1	Requirements – Maintainability	The battery shall be fully charged prior to each test run.	1) Charge until indicator shows "charged/full." 2) Record state of charge before running; if below threshold, recharge.	Operator	Charger indicator photo; SOC note.
3.1.1	Requirements – Maintainability	The battery pack shall be chargeable with the lab charger and indicate charging status.	1) Connect charger; charging indicator appears within 1 min. 2) Observe charger voltage/current values as expected. 3) Connector is secure and polarity is correct.	Operator	Charger display photo; checklist.
3.2	Requirements – Maintainability	Mechanical joints shall be tightened if loose; all screws, nuts and bolts shall be of appropriate sizes and correctly installed.	1) Hand-check all joints; tighten any looseness. 2) Visual check shows zero mismatched fasteners. 3) Post-check confirms no play.	Team	Assembly checklist; photo log.
4.1.1	Requirements – Operability	The operator shall prepare the robot from power-on to ready state within 5 minutes.	1) Start timer at power-on; reach "ready" (nodes up, topics visible) $\leq$5:00. 2) Complete pre-run checklists.	Operator	Timer visible; checklist signed.
4.1.2	Requirements – Operability	The robot shall complete the required tasks within 20 minutes.	1) Start-to-last-placement time $\leq$20:00; timer visible on video. 2) All mandatory steps completed; missing steps = 0.	Team	Video with timer; task checklist.
4.2	Requirements – Operability	The robot shall provide periodic status updates during the run.	1) Status printed/logged at least every 5 s. 2) Messages include state, battery level, and current mode.	Test Eng.	Status log; screenshot.
4.2.1	Requirements – Operability	The robot shall provide status via CLI or GUI while running.	1) Terminal or GUI displays status throughout the run. 2) No gaps >5 s in status updates.	Test Eng.	GUI/CLI screenshot; log.
4.3	Requirements – Operability/Safety	The robot shall detect, and report errors encountered during the process.	1) Inject a representative fault; an error message appears within 2 s. 2) Robot enters safe state (motors stop) until cleared.	Test Eng.	Fault injection video; error log.

Requirement No.	Source	Shall Statement	Verification Success Criteria	Responsible Party?	Agreed Results / Output of test
4.3.1	Requirements – Operability/Safety	The user shall be able to interrupt the process when an error is recognized.	1) Operators stop command halts motion ≤1.0 s. 2) System acknowledges stop in the log and requires explicit resume.	Operator	Video
4.4	Requirements – Operability/Safety	The operator shall mitigate emergency conditions by following the recovery procedure.	1) Follow the safety checklist to reach safe idle within 30 s. 2) No motion permitted until manual resume after checks.	Operator	Safety checklist; recovery video.
4.4.1	Requirements – Operability/Safety	The robot shall include a physical emergency stop button that halts motion and requires manual reset.	1) Press E-stop; robot motion stops within ≤1.0 s. 2) Manual reset required before restarting; documented.	Operator	E-stop video; checklist entry.
4.5	Requirements – Operability	The robot shall be controllable manually.	1) Teleop forward, turn, and stop for 2 minutes without error. 2) No command dropouts or safety violations.	Operator	Teleop video; control log.
4.5.1	Requirements – Operability	The joystick shall be capable of controlling the robot's movement with correct mapping and acceptable latency.	1) Mapping verified for forward/left/right/stop. 2) Command latency ≤0.2 s measured from input to motion.	Test Eng.	Mapping checklist; latency log/video.
5.1	Requirements – Mechanical/ Structural	The robot shall have a chassis that supports system integration and movement	1) Assembly inspection passes; cables are secured. 2) After 10 minutes of driving, no looseness or rubbing was observed. All modules shall be securely mounted and cable-managed.	Team	Team; Assembly checklist; post-run inspection notes.
5.1.1	Requirements – Mechanical/ Structural	The chassis shall support a payload of up to 3 kg without deformation on a flat floor.	1) Mount a 3 kg payload; drive 2 m on flat floor. 2) No structural deformation or scraping during/after testing.	Test Eng.	Video and photos before/after.
5.1.2	Requirements – Mechanical/ Structural	The robot shall use wheels to navigate the arena.	1) Wheel drive propels the robot ≥2 m on flat floor. 2) No excessive slip or derail is observed.	Operator	Video and drive log.
6.1	Requirements – Electrical	The robot shall operate from battery power and from a laboratory power supply.	1) Boot and run from battery; then from lab PSU; both stable. 2) Power source changeover is safe; no resets during the 20min run.	Test Eng.	Photos of power sources; runtime log.

Requirement No.	Source	Shall Statement	Verification Success Criteria	Responsible Party?	Agreed Results / Output of test
6.1.1	Requirements – Electrical	The battery shall provide a regulated 12 V DC output under load.	1) Under ≥1 A load, voltage measures 11–13 V. 2) Polarity corrects; connector secure; ripple within expected bounds.	Test Eng.	Multimeter photo; load note.
6.1.2	Requirements – Electrical	The battery shall provide a minimum runtime of 30 minutes under typical duty.	1) Run a typical duty cycle for ≥30 min without shutdown. 2) Log start/end time and battery SOC; no thermal/low-voltage cutoffs.	Team	Timer-visible video; runtime log.